



An efficient boundary-aware adversarial model for polyp segmentation in colonoscopy images

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ABSTRACT

Accurate and efficient segmentation of colorectal polyps from colonoscopic images is essential for early cancer detection and clinical decision-making. This study presents a compact adversarial segmentation model that integrates edge-guided attention with a refinement-aware discriminator, aiming to improve boundary localization under constrained computational resources. Despite its lightweight architecture, the proposed model outperforms both transformer-based and mobile-efficient baselines across five public datasets. It is worth noting that the method achieves state-of-the-art performance on the Kvasir dataset. It also preserves competitive accuracy in difficult cross-domain settings, where many large-scale models often lose generalization ability. Extensive ablation studies validate the contribution of each architectural component, and inference-time analysis shows tightly bounded latency variation (about 8.5%) even under CPU-only settings. This evaluation shows that the architecture maintains tight execution bounds, enabling reliable integration into latency-sensitive clinical imaging systems operating at 25–30 FPS. These results demonstrate that well-designed adversarial learning can effectively balance accuracy, robustness, and efficiency in biomedical image segmentation. The source code is available at <https://github.com/TongDuyDat/e-Ira>.

1. Introduction

Image segmentation has become an key component in the broader landscape of computer vision (CV), where the task is to assign a categorical label to each pixel of an image, effectively separating distinct objects or structures of interest [1]. While general segmentation techniques have matured considerably in natural images, medical image segmentation remains challenging due to domain-specific issues such as low contrast, high noise, and complex anatomical variability [2].

In many clinical applications, polyp segmentation from colonoscopy images has become an important focus in recent years. This is driven by the heavy clinical burden of colorectal cancer (CRC), which is one of the leading causes of cancer-related deaths worldwide [3]. Polyps are precancerous lesions in the colon that, if detected and removed early, can prevent the progression to CRC [1,2]. Therefore, accurate polyp segmentation is essential for early diagnosis, timely intervention, and improved patient prognosis [4].

To address this challenge, the research community has proposed a variety of deep learning-based methods, ranging from classical CNN architectures such as U-Net [5] and U-Net++ [6], to multi-scale attention mechanisms (e.g., MSRA [7]) and transformer-based networks (e.g., Polyp-PVT [4]). These methods have shown good performance

on public datasets such as CVC-ClinicDB and Kvasir-SEG. Their success comes from the ability to learn hierarchical features and capture global context.

Nevertheless, significant challenges remain. One of the most persistent issues is the high variability in polyp morphology, including shape, size, and texture, which directly impacts the accuracy of segmentation models [2,8]. In addition, colonoscopy images frequently contain noise and artifacts, such as specular highlights, bubbles, and blood, further complicating the segmentation process [2]. Transformer-based architectures, while powerful in handling global context, often demand substantial computational resources, making their integration into real-time clinical settings challenging [1,2].

To overcome these limitations, recent research has explored a range of refinement techniques. These include the integration of boundary-aware modules and edge-guided attention mechanisms to better preserve polyp boundaries and reduce false positive/negative predictions [2,6,9]. However, most of these efforts still rely on either heavy-weight architectures or post-processing steps that are not seamlessly integrated into the end-to-end training pipeline.

Motivated by these challenges, in this study, we propose a novel, lightweight, and fully learnable boundary refinement framework that

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directly addresses the issues of polyp variability and computational burden. Our approach introduces a new discriminator refinement module, termed *Lightweight Refinement Attention (LRA)*, which integrates edge-guided attention maps and depthwise separable convolutions in a fully differentiable manner. The design ensures that boundary information is effectively incorporated while maintaining a low computational footprint. This improves both the robustness and practicality of the system in real clinical workflows. The contributions of this study are summarized as follows:

1. A novel edge-guided lightweight refinement model has been developed for enhancing the discriminator in GAN-based polyp segmentation, with a focus on improved boundary segmentation and computational efficiency.
2. The proposed refinement module leverages depthwise separable convolutions and squeeze-and-excitation attention mechanisms to integrate edge information directly extracted from input images.
3. Comprehensive experiments have been conducted on multiple publicly available polyp segmentation datasets to assess the effectiveness of the proposed approach.

This work bridges the gap between lightweight model design and accurate boundary-aware segmentation. It marks an important step forward in automated polyp segmentation and provides both scientific and practical value for computer-aided diagnosis in colonoscopy.

The remainder of this paper is organized as follows. Section 2 presents a review of related work in polyp segmentation, covering convolutional neural network (CNN)-based architectures, attention mechanisms, and transformer-based frameworks. In Section 3, the proposed methodology is described in detail, including the design principles and technical implementation of the lightweight edge-guided refinement module. Section 4 reports experimental evaluations conducted on several benchmark datasets. The final section concludes the paper, summarizing the main findings and outlining future research directions.

2. Related works

Polyp segmentation in colonoscopy images has become a critical task in medical image analysis, directly supporting early diagnosis of colorectal cancer and minimizing invasive interventions. Despite significant progress, challenges such as morphological variability, imaging artifacts, and real-time deployment requirements continue to drive innovations in this field.

Early approaches relied on hand-crafted features such as color, texture, and shape [10–12]. While these methods offered initial insights, their sensitivity to variations in lighting and viewpoint limited their clinical applicability. Consequently, recent research has shifted towards deep learning-based frameworks, which have demonstrated superior performance in addressing the diverse challenges of polyp morphology and imaging artifacts.

2.1. Encoder–decoder architectures

A dominant paradigm for medical image segmentation is the encoder–decoder framework, with U-Net [5] and U-Net++ [6] as foundational models. U-Net exploits skip connections to combine shallow spatial details with deeper semantic features. However, as highlighted by Jha et al. [13], standard skip connections alone cannot sufficiently capture the complex morphology of polyps and are susceptible to gradient degradation in deep architectures. To overcome these limitations, the ResUNet++ architecture integrates residual units to improve gradient flow, as well as squeeze-and-excitation blocks, ASPP modules, and attention blocks to enhance multi-scale feature aggregation and boundary precision [13]. While these additions significantly improve segmentation performance on challenging polyp datasets, they introduce additional parameter overhead, potentially limiting real-time deployment.

2.2. Attention-enhanced and boundary-aware methods

To improve focus on relevant polyp regions and suppress background noise, attention modules have been incorporated. SE blocks [14] adaptively recalibrate channel-wise feature responses, while CBAM [15] fuses channel and spatial attention. In polyp segmentation, attention modules have been shown to improve detection of small or flat polyps, which often lack strong contrast with surrounding mucosa.

Boundary-aware refinements have also been explored, with Lin et al. [16] proposing BSCA-Net, which leverages bit-slice context attention and multi-path fusion modules to enhance boundary feature extraction and improve polyp margin identification. MCA-Net [17] employs a multi-cascade feature aggregation module (MFA) to progressively refine feature representations, selectively fusing relevant cues from different levels and suppressing background noise to improve boundary precision in polyp segmentation. Boundary Refinement Network (BRNet) [18] employs a boundary generation module to create edge maps by integrating spatial and semantic information, and a boundary-guided refinement module that progressively refines polyp features through top-down supervision. While this design enhances boundary accuracy, it increases model complexity and may limit real-time deployment.

Beyond the methods summarized above, several boundary-aware directions have consolidated. BCNet integrates cross-layer attention with bilateral boundary extraction to learn region and edge cues within an encoder–decoder CNN [19]. BU-Net models margin ambiguity by combining a boundary exploration module (BEM) at shallow layers with a BUM under hybrid supervision on a transformer backbone [20].

Boundary guidance has also shown to be effective. BFNet employs deformable-convolution-based feature aggregation and boundary-guided enhancement, achieving practical efficiency at low resolution [21]. BGGL-Net fuses global–local features with Laplacian/LoG boundary cues on a P2T backbone, offering competitive Params/GFLOPs and improved segmentation of small objects [22]. Finally, a large pretrained model transfer has been investigated with SAM-Mamba, where adapters and a Mamba-prior are used to add multi-scale boundary cues into SAM. Although this improves boundary awareness, the high input resolution and reliance on prompts in SAM continue to limit its use in clinical practice [23].

2.3. Transformer-based and hybrid architectures

Transformer-based architectures have changed segmentation research by capturing long-range dependencies that CNNs often fail to model effectively. In polyp segmentation, Polyp-PVT [4] employs a transformer encoder along with dedicated modules such as cascaded fusion and similarity aggregation to address the fusion of cross-level features and improve the representation of polyp boundaries. Similarly, TransFuse [24] combines transformer and CNN branches in a parallel architecture, introducing a BiFusion module that efficiently merges global context with local details. It achieves a robust performance with reduced parameter complexity. These models outperform pure CNN approaches on public datasets such as Kvasir-SEG [25], especially for complex polyp shapes.

Recent work has further extended this paradigm by integrating MultiResUNet with dual Transformer encoders to jointly capture multi-scale features and global context. Combined with deep supervision and Sharpness-Aware Minimization (SAM), such hybrid architectures demonstrate improved generalization across modalities like CT and ultrasound [26]. Moreover, domain adaptation has emerged as a complementary strategy to mitigate cross-center distribution shifts. Notably, MBDA-Net [27] leverages multi-source prototype alignment, boundary-aware learning, and double normalization to enhance segmentation robustness. However, transformers require large-scale pre-training and significant computational resources [2], making them less suitable for deployment in resource-constrained clinical settings.

2.4. Multi-scale and contextual aggregation

Multi-scale contextual feature aggregation has been recognized as an effective strategy to address the inherent variability in polyp size and shape. MSRA-Net [7] employs a residual attention mechanism to integrate complementary information across scales, using multi-modal data fusion and multi-scale attention to enhance boundary segmentation in challenging segmentation scenarios. Similarly, the Feature Pyramid Network (FPN) [28] constructs feature pyramids through a top-down architecture with lateral connections, providing high-level semantic maps at multiple scales with minimal computational overhead. DenseASPP [29] extends this idea by densely connecting atrous convolutional layers with varying dilation rates, enabling more comprehensive multi-scale sampling without significantly increasing model complexity. While these approaches improve the segmentation accuracy of diverse structures by fusing multi-scale features, they often rely on heavy backbones such as ResNet or DenseNet, which can limit real-time performance and complicate integration into lightweight systems.

2.5. Lightweight and real-time models

In semantic segmentation, the pursuit of efficiency has driven the development of compact architectures that lower computational cost while preserving strong feature representations. LiteSeg and DeepLabV3 adopted MobileNetV2 encoders that employ depthwise separable convolutions and inverted residuals, demonstrating that lightweight backbones can deliver good accuracy with far fewer parameters [30,31]. MHC-PSPNet further validated the adaptability of such CNN-based approaches in constrained environments [32]. More recently, SegFormer introduced a Transformer-based alternative that combines hierarchical encoders with lightweight MLP decoders, showing that efficiency can also be achieved beyond purely convolutional designs [33].

In addition, lightweight CNNs have found broad application in medical imaging. MobileNetV2 was integrated into diverse diagnostic tasks, including skin cancer classification [34], brain tumor detection [35], and white blood cell analysis [36]. This reliability makes it well suited for healthcare applications that demand both precision and efficiency.

In colonoscopy, the need for real-time segmentation has encouraged the adaptation of lightweight models for polyp analysis. PolypSeg+ introduced adaptive context modeling with a lightweight feature pyramid fusion to efficiently address scale variation [37]. An improved DeepLabV3+ replaced heavy encoders with MobileNetV2 and simplified pooling operations, thereby reducing complexity while retaining segmentation quality [38]. ColonSegNet emphasized standardization and speed by providing a benchmark framework for detection, localization and segmentation [39]. These studies show the shift from general lightweight segmentation models to specialized frameworks for colonoscopy, where MobileNetV2 often proves to be a reliable and practical backbone.

2.6. Motivation and distinction of this study

Despite significant advancements in efficient backbone architectures and attention-based modules, many lightweight segmentation models still lack explicit mechanisms for improving boundary precision as an essential factor in clinical image interpretation. While methods such as PolypSeg+ [37] and SegFormer [33] offer competitive trade-offs between speed and accuracy, they often do not incorporate dedicated boundary refinement components or depend on non-trainable post-processing stages, which hinders full integration into real-time pipelines. In addition, boundary-aware networks such as BCNet [19], BFNet [21], BGGL-Net [22], BUNet [20], and SAM-Mamba [23] have demonstrated good improvements in edge localization. However, their

dependence on heavy backbones and large computational costs, constrains their applicability in clinical practice where efficiency and robustness are required.

To overcome this limitation, we introduce a novel lightweight refinement module that is directly embedded within the GAN discriminator. Rather than treating edge enhancement as an optional process, our design places boundary modeling at the core of adversarial supervision. It integrates depthwise separable convolutions and squeeze-and-excitation attention to selectively enhance boundary-relevant features with minimal computational burden. This contribution strengthens lightweight segmentation by combining efficiency with precise boundary refinement in a fully end-to-end trainable architecture. As a result, it is suitable for real-time diagnostic applications.

3. Methods

Polyp segmentation plays a crucial role in clinical decision-making and surgical planning, where precise segmentation of polyp boundaries can directly influence treatment outcomes. However, achieving high segmentation accuracy while maintaining real-time inference speed remains a challenge in clinical endoscopy applications [2]. This study proposes a novel lightweight adversarial segmentation framework that incorporates an edge-guided refinement attention mechanism to address these limitations.

The proposed framework is built upon a generative adversarial network (GAN) paradigm, comprising two main components: a lightweight generator for mask prediction and an edge-guided refinement discriminator for adversarial supervision, as shown in Fig. 1. This structure maintains an end-to-end trainable pipeline, facilitating efficient deployment in clinical scenarios.

3.1. Generator

Accurate and efficient polyp segmentation is essential for reliable colorectal cancer screening. To achieve this, the generator $\mathcal{G}$ is constructed as a lightweight encoder-decoder network, leveraging MobileNetV2 [40,41] as the backbone. This choice is motivated by its proven capability in reducing computational overhead while maintaining strong feature expressiveness as critical factors for real-time clinical deployment in colonoscopy procedures [37].

Input and encoder feature extraction: Let $I \in \mathbb{R}^{H \times W \times 3}$ denote the input colonoscopy image. The encoder $\mathcal{E}$ extracts a hierarchy of feature maps (Eq. (1)):

$$F_e = \mathcal{E}(I), \quad (1)$$

where $F_e \in \mathbb{R}^{\frac{H}{s} \times \frac{W}{s} \times C}$, s denotes the spatial downsampling factor, and C represents the number of channels in the deepest encoder layer.

Depthwise separable convolutions: The encoder primarily utilizes depthwise separable convolutions—an operation that factorizes a standard convolution into a depthwise spatial convolution and a pointwise convolution, significantly reducing computational complexity from $O(K^2MN)$ to $O(K^2M + MN)$ [42] (Eq. (2)):

$$\text{DWConv}_{3 \times 3}(X) = \text{PW}(\text{DW}(X)), \quad (2)$$

where $\text{DW}(\cdot)$ represents a 3×3 depthwise convolution, and $\text{PW}(\cdot)$ denotes a 1×1 pointwise convolution. This decomposition is critical for achieving low-latency inference while still keeping detailed spatial information. This is especially necessary for endoscopic video frames, where polyps can appear in many shapes [9].

Decoder and multi-scale feature fusion: The decoder $\mathcal{D}$ is designed to progressively upsample and fuse encoder features, ensuring robust recovery of spatial resolution and boundary details (Eq. (3)):

$$F_d^l = \text{Up}(F_d^{l+1}) + F_e^l, \quad (3)$$

where $\text{Up}(\cdot)$ denotes bilinear interpolation for upsampling, and F_e^l represents encoder features at level l . This skip connection-based fusion

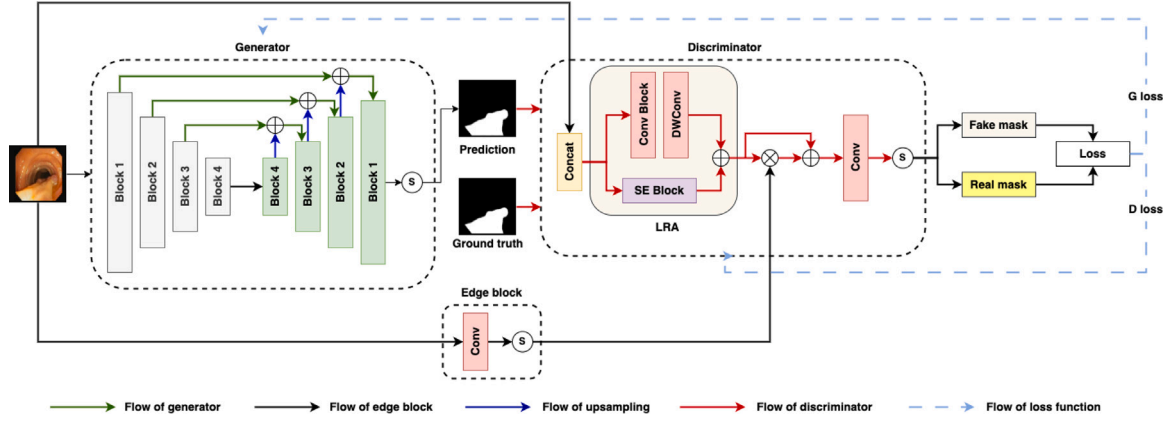


Fig. 1. The overview of our proposed architecture, which contains some key components.

preserves high-frequency boundary information, which is essential for precise identification of polyps, particularly in low-contrast images [10, 11].

Segmentation output: The final segmentation mask is computed as Eq. (10):

$$M_{\text{pred}} = \sigma(D(F_d)), \quad (4)$$

where $\sigma(\cdot)$ is the sigmoid activation function, yielding $M_{\text{pred}} \in [0, 1]^{H \times W \times 1}$. The output M_{pred} serves both as the clinical segmentation prediction and as an adversarial input to the discriminator D_{adv} for adversarial training.

Design justification: This encoder–decoder generator architecture ensures compatibility with the edge-guided refinement attention discriminator. The use of depthwise separable convolutions and skip-connected feature fusion shows a clear effort to keep computation low while preserving spatial detail. This design helps address the dual challenges of morphological variability and real-time processing in clinical settings. Importantly, the generator is fully end-to-end differentiable, which fits well with the adversarial training pipeline. This allows joint optimization of spatial coherence and boundary precision in polyp segmentation.

3.2. Discriminator

To address the limitations of conventional post-processing methods and to enhance boundary-aware discrimination within the adversarial framework, a novel *Lightweight Refinement Attention (LRA)* discriminator is proposed. In contrast to computationally demanding Conditional Random Fields (CRF) [43], the proposed discriminator directly integrates refinement into the adversarial learning process, maintaining end-to-end differentiability and practical deployment feasibility.

The discriminator receives a composite input formed by concatenating the raw colonoscopy image $I \in \mathbb{R}^{H \times W \times 3}$ with the predicted segmentation mask $M_{\text{pred}} \in \mathbb{R}^{H \times W \times 1}$. This forms a four-channel tensor Eq. (5):

$$x_{\text{in}} = \text{Concat}(I, M_{\text{pred}}) \in \mathbb{R}^{H \times W \times 4}. \quad (5)$$

Such composition allows the discriminator to jointly consider both appearance-based cues and spatial-semantic consistency as an essential requirement for accurate polyp boundary evaluation [2].

Hierarchical feature representations are extracted from x_{in} using five convolutional blocks, each employing a 3×3 convolution, stride 2 downsampling, and LeakyReLU activation ($\alpha = 0.2$). This hierarchy progressively compresses spatial resolution while increasing channel dimensionality, yielding a compact representation Eq. (6):

$$F = C_5(\dots C_2(C_1(x_{\text{in}}))), \quad (6)$$

where $C_i(\cdot)$ denotes the i th convolutional block.

To enhance the discriminator's focus on polyp boundaries, an edge attention map $A_e \in \mathbb{R}^{H \times W \times 1}$ is generated directly from the input image Eq. (7):

$$A_e = \sigma(\text{Conv}_{3 \times 3}(I)), \quad (7)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. This edge attention emphasizes regions of structural significance, helping to suppress background noise and improve boundary sensitivity. The attention map is resized via bilinear interpolation to match the spatial dimensions of F .

The core of the discriminator's refinement lies in the proposed LRA module. Motivated by the efficiency of depthwise separable convolutions [42] and the channel recalibration of squeeze-and-excitation (SE) blocks [14], the refinement process is formulated as Eq. (8):

$$F' = \text{DWConv}_{3 \times 3}(F) + \text{SE}(F), \quad (8)$$

where $\text{DWConv}_{3 \times 3}(\cdot)$ captures localized spatial information while $\text{SE}(\cdot)$ adaptively reweights channel importance. The SE block enhances feature representation by adjusting channel responses. It reduces the influence of irrelevant signals while reinforcing those linked to boundary cues. This block allows the discriminator to effectively capture boundary-sensitive features and improves its ability to distinguish sharp, consistent edges from blurred ones. In adversarial training, the SE block also shapes the feedback to the generator, encouraging the production of masks with cleaner and more accurate boundaries. Compared with heavier attention mechanisms such as CBAM [15] or Transformer-based modules [4], SE offers a lightweight yet effective solution well suited for our model.

The refined feature map F' is further modulated by the edge attention A_e through element-wise multiplication and a residual connection Eq. (9):

$$F_{\text{refined}} = A_e \odot F' + F', \quad (9)$$

where $\odot$ denotes element-wise multiplication. This residual-attention fusion enhances the network's capability to detect subtle polyp boundaries while preserving spatial coherence.

The final patch-wise probability map is computed by applying a 1×1 convolution followed by a sigmoid activation to F_{refined} Eq. (10):

$$D(x) = \sigma(\text{Conv}_{1 \times 1}(F_{\text{refined}})), \quad (10)$$

indicating the discriminator's confidence in distinguishing real from generated segmentation masks and providing localized feedback during adversarial training.

This edge-guided LRA-based design ensures that boundary refinement is fully learnable and directly optimized within the adversarial

loop unlike conventional CRF-based refinements that rely on separate post-processing [43]. As a result, the discriminator not only achieves stronger boundary localization and spatial coherence but also significantly reduces computational overhead as key considerations for real-time clinical deployment. The end-to-end nature of the refinement process further ensures robust generalization across varied polyp morphologies and imaging artifacts [2].

3.3. Loss function

The performance of a GAN-based segmentation framework strongly depends on the design of its loss function, which must balance pixel-level accuracy, structural coherence, and adversarial consistency [44, 45]. In this work, a hybrid loss formulation is employed that integrates a multi-term segmentation loss with an adversarial objective, enabling the generator to produce high-quality masks with precise boundaries and realistic topologies.

Segmentation loss: to address the challenges of class imbalance, fine-grained boundaries, and topological irregularities often found in colorectal polyp images, the segmentation loss $\mathcal{L}_{\text{seg}}$ is constructed as a combination of three complementary components as in Eq. (11): binary cross-entropy (BCE) Eq. (12), Dice loss Eq. (13), and intersection-over-union (IoU) loss Eq. (14):

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{IoU}}. \quad (11)$$

Each term captures a distinct perspective of mask quality:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (12)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i + \epsilon}, \quad (13)$$

$$\mathcal{L}_{\text{IoU}} = 1 - \frac{\sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N (y_i + \hat{y}_i - y_i \hat{y}_i) + \epsilon}. \quad (14)$$

Here, y_i and $\hat{y}_i$ denote the ground-truth and predicted values at pixel i , and ϵ is a small constant added for numerical stability. While Dice loss promotes region-level overlap, it may not sufficiently penalize over-segmentation or missed thin structures. IoU loss complements Dice by enforcing stricter shape conformity, particularly in irregular polyp contours. The combination of BCE, Dice, and IoU thus provides both local pixel precision and global boundary alignment.

Adversarial objective: to further enhance mask realism and structural consistency, the model adopts an adversarial learning strategy in which a discriminator D is trained to distinguish between predicted and real masks conditioned on the input image I . The generator G is concurrently trained to produce masks that fool the discriminator. The adversarial loss is defined as Eq. (15):

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{I, M_{\text{real}}} [\log D(I, M_{\text{real}})] + \mathbb{E}_I [\log (1 - D(I, G(I)))], \quad (15)$$

The overall training objective follows a minimax game between the generator and discriminator Eq. (16):

$$\min_G \max_D \mathcal{L}_{\text{adv}}(G, D), \quad (16)$$

where the discriminator is optimized using the binary classification loss Eq. (17):

$$\mathcal{L}_D = -\mathbb{E}_{I, M_{\text{real}}} [\log D(I, M_{\text{real}})] - \mathbb{E}_I [\log (1 - D(I, G(I)))] \quad (17)$$

and the generator is updated using both segmentation and adversarial feedback Eq. (18):

$$\mathcal{L}_G^{\text{adv}} = -\mathbb{E}_I [\log D(I, G(I))]. \quad (18)$$

The final loss function used to update the generator is a linear combination of segmentation and adversarial terms Eq. (19).

$$\mathcal{L}_G^{\text{total}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_G^{\text{adv}}, \quad (19)$$

In practice, the training alternates between updating D and G , allowing the generator to improve structural fidelity and generalization robustness while preserving segmentation accuracy.

4. Experiments

4.1. Datasets

To evaluate the effectiveness and generalization capability of the proposed model, we conducted extensive experiments on five publicly available and widely adopted benchmark datasets for polyp segmentation: Kvasir-SEG [25], CVC-ClinicDB [46], CVC-ColonDB [11], CVC-T [47], and ETIS-LaribPolypDB [48]. These datasets have been selected due to their diversity in image resolutions, polyp morphology, and acquisition environments, offering a comprehensive assessment across both high- and low-quality endoscopic images.

Kvasir-SEG [25] contains 1000 colonoscopy images with pixel-level annotations, covering a wide range of polyp sizes, shapes, and illumination conditions. It is among the most frequently used datasets in recent studies due to its rich variability and balanced sample size.

CVC-ClinicDB [46] includes 612 frames extracted from colonoscopy videos, each paired with a manually annotated binary mask. This dataset is known for its relatively low resolution (384×288) and serves as a practical benchmark for testing models under constrained visual quality.

CVC-ColonDB [11] consists of 380 higher-resolution colonoscopic images (574×500), offering clear delineation of polyp structures, particularly useful for testing fine boundary detection.

CVC-T [47] is a smaller dataset with only 60 images but remains relevant due to its challenging lighting and polyp localization conditions.

ETIS-Larib [48] comprises 196 high-resolution images (1225×966), presenting complex polyp backgrounds and significant inter-image variability. It is considered one of the most difficult datasets and thus is ideal for validating model robustness in adverse settings.

These datasets have been extensively used in prior work such as [8, 9, 49] establishing a common ground for fair comparison among state-of-the-art methods.

4.2. Protocols and metrics

4.2.1. Protocols

To ensure fair and reproducible evaluation of the proposed method, we adopted several experimental protocols commonly used in recent polyp segmentation studies [8, 49]. These protocols aim to assess both the performance on familiar data distributions and the generalization capability across datasets.

Following previous studies, we used a standard split where 80% of the images from the Kvasir and CVC-ClinicDB datasets were used for training, 10% for validation, and the remaining 10% along with the complete sets of CVC-ColonDB, CVC-T, and ETIS-Larib were used for testing. This allows for direct comparison under unified benchmark settings.

Five-fold cross-validation: to further reduce the risk of overfitting and provide a more robust estimation of performance, we conducted five-fold cross-validation on both Kvasir and CVC-ClinicDB. In each fold, four parts were used for training and one for validation. This strategy provides better insights into model variance and stability.

Cross-dataset evaluation: To simulate real-world deployment scenarios where test data may differ in distribution, a cross-dataset evaluation protocol was adopted [37]. In this setting, the model was trained on one dataset (or a combination) and tested on a completely unseen dataset. For example, the model trained on Kvasir + CVC-ColonDB was evaluated on ETIS-Larib to observe generalizability.

Table 1

Performance comparison with other models. We have adapted these methods into our configuration.

Methods	Kvasir		CVC-ClinicDB		CVC-ColonDB		CVC-T		ETIS-Larib	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
UNet [5]	80.9	73.8	81.5	75.6	53.2	46.1	71.4	63.8	40.2	34.5
UNet++ [50]	83.6	75.9	80.8	74.2	49.2	42.3	72.3	63.8	41.2	35.6
SFA [51]	71.8	60.4	68.6	59.9	48.7	36.3	31.1	23.5	44.9	31.7
HarDNet-MSEG [52]	90.8	84.9	92.7	87.4	74.4	64.3	87.2	81.5	66.3	60.1
TransUNet [53]	91.0	85.5	93.3	88.5	77.9	69.5	89.1	82.1	72.8	65.7
TransFuse-L [24]	91.8	86.7	94.0	89.4	77.8	70.2	89.1	82.3	73.5	66.0
PraNet [8]	88.8	83.2	91.2	86.5	64.2	58.2	87.2	80.7	49.3	45.1
CaraNet [54]	90.2	84.8	91.3	86.5	73.1	66.2	88.6	81.8	66.2	59.4
ColonFormer-S [49]	85.6	78.3	96.7	93.7	77.8	69.6	87.9	80.9	73.7	65.1
PolyPooling-B [9]	93.2	88.6	94.0	90.4	82.7	72.6	91.1	85.9	87.8	80.8
Pix2pix-cGAN [55]	77.2	68.0	68.5	62.8	61.7	52.8	73.9	65.4	56.2	54.2
Pix2pix-cGAN [55] (+ Edge, + LRA)	80.4	71.0	72.2	66.5	64.6	55.7	76.1	67.6	59.1	56.7
Li-DeepLabV3+ [38]	87.7	78.9	85.3	83.9	76.5	67.2	81.3	79.9	61.1	62.2
OURS	95.94	92.53	<u>86.47</u>	<u>80.34</u>	79.38	69.9	91.3	<u>83.6</u>	<u>70.4</u>	<u>69.4</u>

4.2.2. Training settings

All experiments were conducted using the PyTorch framework and executed on a single NVIDIA GeForce RTX 3050 GPU. The training was performed for 200 epochs with a batch size of 16. Input images were uniformly resized to a resolution of 256×256 pixels to maintain consistency across datasets.

The Adam optimizer was utilized due to its adaptive learning rate capability and robustness in gradient-based optimization. The learning rate was initialized at lr_0 and decayed using a step-based schedule. Moreover, the learning rate at epoch e was updated, as mentioned in [56], following Eq. (20):

$$lr_e = lr_0 \times \gamma^{\lfloor \frac{e}{T} \rfloor} \quad (20)$$

where lr_0 denotes the initial learning rate, γ is the decay factor, T represents the interval of epochs after which the learning rate is reduced, and $\lfloor \cdot \rfloor$ is the floor function.

To improve model robustness and generalization, extensive data augmentation techniques were applied during training. These included random horizontal and vertical flips ($p = 0.5$), rotations of 90° , 180° , or 270° ($p = 0.5$), geometric distortions such as elastic, grid, and optical transformations ($p = 0.3$), as well as color operations (adjustments to brightness, contrast, saturation, and hue, $p = 0.3$).

To prevent overfitting and improve generalization, an early stopping mechanism was adopted with a patience threshold of 50 epochs. Training was terminated if the validation performance showed no improvement for 50 consecutive epochs.

4.2.3. Metrics

To evaluate the performance of the proposed segmentation framework, we employ some commonly used metrics: Dice Coefficient (Dice), Intersection over Union (IoU), Precision, and Recall. These metrics provide a robust measure of both pixel-wise accuracy and overlap between the predicted and ground truth masks.

The dice quantifies the overlap between the predicted mask P and the ground truth mask G , and is defined as Eq. (21):

$$\text{Dice} = \frac{|P \cap G|}{|P| + |G|} \quad (21)$$

A higher DSC indicates better spatial agreement between the segmentation and the ground truth.

The IoU also known as the Jaccard Index, measures the ratio between the intersection and the union of the predicted and ground truth regions as Eq. (22):

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \quad (22)$$

The precision reflects the proportion of correctly predicted positive pixels out of all pixels predicted as positive as Eq. (23):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (23)$$

where TP and FP denote true positives and false positives, respectively.

The recall indicates the proportion of ground truth positive pixels that were correctly predicted as Eq. (24):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (24)$$

with FN representing false negatives.

Each of the above metrics reflects a distinct aspect of segmentation quality. When evaluated together, they provide a comprehensive view of the model's strengths and limitations.

4.3. Evaluation

4.3.1. Experimental results

Table 1 presents a performance comparison across five benchmark datasets. The proposed model achieves the best results on the Kvasir dataset, with a mean Dice of 95.94% and mean IoU of 92.53%, significantly outperforming both heavy transformer-based architectures (e.g., TransUNet, TransFuse-L) and lightweight models such as Li-DeepLabV3+ (87.7% / 78.9%).

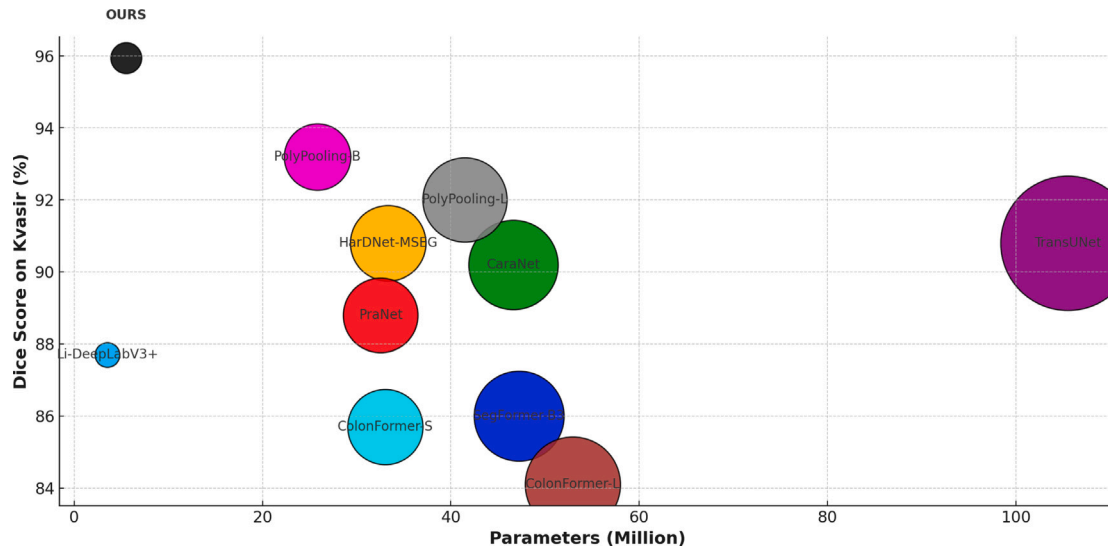
On CVC-ColonDB, a challenging dataset with low contrast and visual noise, our method again leads with 79.38% Dice and 69.9% IoU, demonstrating strong robustness to degraded image quality and irregular polyp structures. For CVC-T and ETIS-Larib, which are known for complex boundaries and small polyp instances, our model maintains high accuracy (91.3% and 70.4% Dice), ranking among the top performers.

While ColonFormer-S reports the highest Dice on CVC-ClinicDB (96.7%), our model achieves a competitive 86.47% Dice and 80.34% IoU without relying on large-scale pretraining or transformer-heavy encoders. Compared to Li-DeepLabV3+ (85.3% / 83.9%), our method

Table 2

Number of parameters and GFLOPs of the different state-of-the-art models.

Method	Parameters (M)	GFLOPs
PraNet [8]	32.55	13.11
HarDNet-MSEG [52]	33.34	11.38
CaraNet [54]	46.64	21.69
TransUNet [53]	105.5	60.75
SegFormer-B3 [33]	47.22	33.68
ColonFormer-S [49]	33.04	16.03
ColonFormer-L [49]	52.94	22.94
PolyPooling-B [9]	25.81	34.77
PolyPooling-L [9]	41.47	40.83
Li-DeepLabV3+ [38]	3.54	–
Ours	5.54	3.61
	Generator Parameters: 1.06	Generator GFLOPs: 1.35
	Discriminator Parameters: 4.47	Discriminator GFLOPs: 2.26

**Fig. 2.** Trade-off between model accuracy and model complexity, measured by the number of parameters.

demonstrates more consistent performance across all datasets, indicating a better balance between segmentation accuracy, generalization, and architectural efficiency.

In addition, Table 1 also presents Pix2pix-cGAN and its variant enhanced with Edge and LRA. The baseline records moderate accuracy (77.2% Dice on Kvasir), while adding Edge and LRA yields consistent gains of about 3%–4% across datasets.

Table 2 summarizes the model complexity of recent state-of-the-art approaches. Transformer-based models like TransUNet and SegFormer-B3 require over 47M parameters and up to 60.75 GFLOPs, posing challenges for real-time or edge deployment.

Even efficient models such as HarDNet-MSEG and ColonFormer-S exceed 11 GFLOPs, indicating room for further optimization. Li-DeepLabV3+, a lightweight MobileNetV2-based model, reduces the parameter count to 3.54M, but lacks adversarial refinement and explicit boundary modeling.

In contrast, our model adopts a compact edge-aware GAN, distributing computation between a lightweight generator (1.06M, 1.35 GFLOPs) and a small discriminator (4.47M, 2.26 GFLOPs), totaling 5.54M parameters and 3.61 GFLOPs. This structure supports adversarial learning while maintaining real-time suitability.

While high-performing models like PolyPooling-L and CaraNet emphasize accuracy through architectural complexity, their large computational costs (up to 40+ GFLOPs) limit use in low-resource environments.

Fig. 2 illustrates the trade-off between segmentation accuracy (Dice score on Kvasir) and model complexity, measured by the number of

parameters. Each circle's area is proportional to the model's parameter count, providing a visual cue of computational scale. Our model (OURS) achieves the highest accuracy while maintaining a compact architecture, occupying the top-left region of the plot. In contrast, several high-performing models such as TransUNet and SegFormer incur significantly larger parameter overheads. Lightweight baselines like Li-DeepLabV3+ offer minimal complexity but fall short in accuracy. This visualization highlights the superior balance of accuracy and model efficiency achieved by our approach.

Table 3 presents the results of five-fold cross-validation on the Kvasir and CVC-ClinicDB datasets. While models such as ColonFormer-S and CRCNet deliver slightly higher mDice or mIoU scores, they require substantially larger architectures. In contrast, our proposed model achieves very competitive performance with 89.8% mDice and 90.3% recall on Kvasir, and 90.1% mDice with 92.7% precision on CVC-ClinicDB, all with a significantly lower computational cost. These results validate the robustness and consistency of our lightweight adversarial design, especially in terms of balanced recall and precision, which are critical for minimizing false negatives in clinical scenarios. To make the comparison with adversarial frameworks clearer, we also include Pix2pix-cGAN as a baseline. The model on its own gives weak results. When Edge and LRA are added, the accuracy improves on both datasets, showing that these modules are useful even for older adversarial models.

To further evaluate the robustness and transferability of the proposed model, we conducted cross-dataset experiments in which the training and testing datasets are strictly disjoint. As shown in Table 4,

Table 3

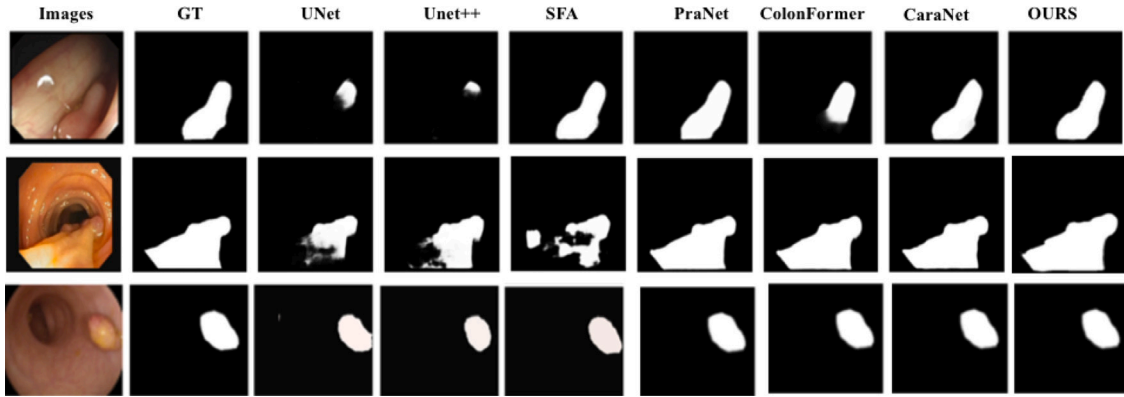
Performance comparison of different methods on five-fold cross-validation of the CVC-ClinicDB and Kvasir datasets.

Methods	Kvasir				CVC-ClinicDB			
	mDice	mIoU	Recall	Precision	mDice	mIoU	Recall	Precision
ResUNet++ [13]	77.4	67.7	82.8	80.5	80.9	74.2	82.7	83.5
DoubleUNet [57]	87.2	80.8	90.8	88.7	91.5	86.2	91.3	92.2
ColonSegNet [39]	67.0	55.2	72.5	72.2	81.1	87.9	91.7	92.4
HarDNet-MSEG [52]	88.2	82.3	89.6	91.9	91.7	88.0	91.8	94.2
PraNet [8]	87.7	81.4	90.5	89.9	92.5	87.9	93.5	92.8
ColonFormer-S [49]	91.6	88.2	94.7	92.0	94.1	89.5	96.3	93.4
CRCNet [58]	91.0	89.7	89.5	91.3	95.4	93.7	93.9	94.7
Pix2pix-cGAN [55]	74.5	65.5	76.8	74.0	66.0	58.0	70.9	63.1
Pix2pix-cGAN [55] (+ Edge, + LRA)	77.3	68.2	79.4	76.6	69.2	61.0	73.8	65.9
OURS	89.8	82.9	90.3	90.5	90.1	88.2	90.8	92.7

Table 4

Performance comparison of different methods on cross-dataset configurations.

Train	Test	Methods	mDice	mIoU	Recall	Precision
CVC-ColonDB + ETIS-Larib	CVC-ClinicDB	ResUNet++ [13]	40.2	29.9	47.6	50.1
		ColonSegNet [39]	43.1	32.6	52.4	54.6
		DoubleUNet [57]	74.3	64.5	76.5	81.7
		HarDNet-MSEG [52]	75.9	67.4	77.9	85.5
		PraNet [8]	77.1	68.1	82.5	81.9
		ColonFormer-S [49]	84.5	76.4	85.9	88.8
		ColonFormer-L [49]	85.4	76.8	83.6	89.6
		OURS	86.2	78.3	86.2	88.3
CVC-ColonDB	CVC-ClinicDB	ResUNet++ [13]	34.2	24.5	38.3	47.9
		DoubleUNet [57]	43.7	38.0	41.9	63.4
		ColonSegNet [39]	57.7	27.0	50.6	46.5
		HarDNet-MSEG [52]	71.6	62.9	73.7	82.4
		PraNet [8]	74.2	65.1	74.5	82.6
		ColonFormer-S [49]	82.2	72.8	81.4	87.4
		ColonFormer-L [49]	79.8	72.6	78.9	88.3
		OURS	79.9	71.3	79.4	85.2
CVC-ClinicDB	ETIS-Larib	ResUNet++ [13]	20.9	15.7	30.5	20.5
		ColonSegNet [39]	22.0	10.9	64.9	14.2
		DoubleUNet [57]	59.4	49.6	69.6	60.5
		HarDNet-MSEG [52]	66.3	58.9	67.0	69.8
		PraNet [8]	62.5	54.9	75.5	60.2
		ColonFormer-S [49]	71.6	63.5	78.9	72.4
		ColonFormer-L [49]	75.3	66.8	86.7	74.0
		OURS	74.2	67.5	82.1	75.8

**Fig. 3.** Visual comparison between our proposed model and some state-of-the-art models.

our model achieves strong performance across all three cross-domain configurations, often surpassing or matching state-of-the-art methods that rely on heavier architectures.

When trained on CVC-ColonDB and ETIS-Larib, our model achieves 86.2% mDice and 78.3% mIoU on CVC-ClinicDB, outperforming all baselines including ColonFormer-L. Similarly, in the CVC-ColonDB and

CVC-ClinicDB setting, our model reaches 79.9% mDice, closely approaching ColonFormer-S (82.2%) while maintaining a significantly smaller model size and computational footprint.

In the most challenging setting as training on CVC-ClinicDB and testing on ETIS-Larib, our method achieves 74.2% mDice and 67.5% mIoU, demonstrating superior boundary localization and generalization

Table 5

Influence of different components on the performance.

Method	Kvasir		CVC-ClinicDB		CVC-ColonDB		CVC-T		ETIS-Larib	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
OURS w/o Edge block	82.5	76.5	74.9	68.0	67.4	58.6	78.3	70.1	63.6	56.1
OURS w/o LRA	81.6	75.7	73.5	66.9	65.2	56.0	77.6	69.3	62.0	54.8
OURS	95.94	92.53	86.47	80.34	79.38	69.9	91.3	83.6	70.4	69.4
OURS (SE→ CBAM [15])	92.4	90.8	84.2	78.4	75.1	67.8	89.7	80.1	68.2	65.7
OURS (SE→ ECA [59])	91.6	89.5	81.6	75.2	75.2	64.1	85.3	77.9	65.6	64.4

under distribution shift. These results confirm that the proposed architecture not only performs competitively in in-distribution settings but also generalizes well across diverse clinical data sources.

Fig. 3 compares our proposed model with other state-of-the-art methods. Our model demonstrates highly competitive performance, producing segmentation masks with smooth and accurate boundaries. In various cases, especially those involving small or low-contrast polyps, it captures object shapes more precisely than other methods. These results highlight the effectiveness of our approach in generating clean and reliable segmentation outputs.

4.3.2. Ablation studies

To better understand the contribution of key components within our proposed framework, we perform ablation studies focusing on two central modules: the LRA block in the discriminator and the edge-guided attention block derived from the input image.

As summarized in Table 5, removing either component leads to a notable degradation in performance across all datasets. Specifically, removing the LRA block causes a significant drop in Dice and IoU scores, ranging from 15% to 25% highlighting its importance in refining spatial structure and enforcing boundary-level adversarial consistency. Without LRA, the discriminator fails to adequately guide the generator in resolving complex and subtle polyp boundaries, especially under noisy or low-contrast conditions.

Similarly, excluding the edge-guided attention block results in a 10%–15% decrease in segmentation accuracy. This module provides a learnable structural prior extracted directly from the input image, enhancing the model's ability to localize boundary transitions and suppress background artifacts. Its removal leads to more diffused predictions, especially around polyp contours. These results confirm that both the LRA module and edge-guided block play a crucial role in achieving high-precision lightweight segmentation.

To further examine the role of the SE block in the LRA module we compared it with two alternative attention mechanisms CBAM [15] and ECA [59]. The results in Table 5 show that CBAM performs slightly worse than SE in terms of mDice and mIoU. In our setting the additional number of parameters and FLOPs is not very large because the discriminator contains only one attention block with a limited number of channels, but CBAM still introduces extra structural complexity due to the spatial attention branch which reduces its efficiency for our context. ECA keeps a compact design similar to SE, but its accuracy is lower, which suggests that it cannot capture boundary information effectively. These observations confirm that SE provides the best balance between accuracy and efficiency and therefore is chosen for the LRA design.

Fig. 4 provides a comparative view of inference-time distributions across three deployment variants. In real-time image processing systems, predictability defined as the bounded variance around mean execution time is often prioritized over raw speed, as it ensures temporal determinism and schedulability under frame-rate constraints [60]. The full proposed model demonstrates superior temporal stability, with less than $\pm 8.5\%$ deviation around the mean (approximately 59ms), indicating high consistency across frames and safeguarding against frame drops or buffer underflow. While eliminating the edge block slightly narrows the spread, it potentially sacrifices boundary-sensitive segmentation fidelity. On the other hand, removing the LRA makes the processing time less stable, which reduces the system's predictability.

Table 6

Effect of individual augmentation techniques on the Kvasir-SEG dataset. The results are reported in terms of mDice and mIoU.

Setting	mDice	mIoU
All augmentations (OURS)	95.94	92.53
Brightness only	93.60	90.20
Contrast only	93.35	90.05
Hue only	93.72	90.30
Saturation only	93.20	89.85
Flip only	94.05	90.70
Rotation only	94.18	90.85
Elastic deformation only	93.95	90.50
Grid distortion only	94.25	90.90
Optical distortion only	93.80	90.40

In practice, real-time systems demand that processing times fall within narrow, repeatable intervals to support pipeline-level synchronization and avoid deadline violations [61]. Crucially, these results were obtained during inference on a standard CPU configuration as an AMD Ryzen™ 5 5600H processor (3.30 GHz up to 4.20 GHz, 16 MB cache) paired with 16 GB RAM without GPU acceleration. This evaluation proves that our architecture maintains tight execution bounds, making it suitable for integration into latency-sensitive clinical imaging systems at 25–30 FPS.

Moreover, as reported in Table 6, applying each augmentation individually results in performance within about 2%–3% of the our setting that combines all augmentations. Geometric operations including grid distortion, rotation, and flipping perform slightly better than appearance-based adjustments such as brightness, contrast, hue, and saturation. This highlights the value of structural variation in polyp segmentation. Elastic and optical distortions also offer minor improvements, though they remain below the combined strategy. The highest accuracy is achieved with the full augmentation strategy (mDice 95.94, mIoU 92.53). It informs that diversity in augmentation enhances robustness, while the major improvements of our framework are mainly driven by the proposed LRA design.

4.3.3. Discussions

The proposed method shows a better balance between segmentation accuracy, computational efficiency, and deployment feasibility. These three aspects are core requirements for real-time medical image segmentation systems. Moreover, the observed performance is closely linked to architectural design and optimization strategies.

High segmentation accuracy on datasets such as Kvasir and CVC-ColonDB is primarily enabled by the effective integration of the edge-guided attention block and the LRA module. These components enhance structural sensitivity and boundary precision, particularly beneficial in detecting small or low-contrast polyps under challenging visual conditions.

In cross-dataset experiments, the method shows strong generalization ability. When trained on CVC-ColonDB and tested on CVC-ClinicDB, the model yields competitive performance comparable to transformer-based architectures, despite relying on significantly fewer parameters and limited pretraining. Even in the most adverse scenario with training on CVC-ClinicDB and testing on ETIS-Larib, the model

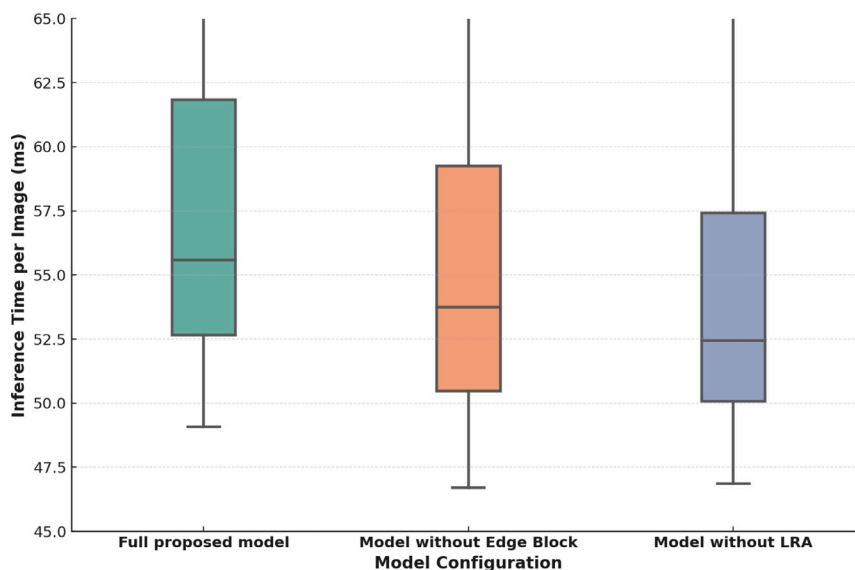


Fig. 4. Distribution of execution time across model variants. All measurements were performed on a CPU-only setup using an AMD Ryzen™ 5 5600H processor (3.30 GHz up to 4.20 GHz, 16 MB Cache) with 16 GB RAM, without GPU acceleration.

preserves high Dice and IoU scores. This shows strong robustness to domain shift and structural variation.

The impact of core components is further validated through ablation studies. Exclusion of the LRA module results in a notable decline in Dice and IoU (15%–25%), underscoring its role in spatial refinement and adversarial guidance. Similarly, removing the edge-guided module leads to 10%–15% accuracy loss, demonstrating its importance in enhancing contour awareness and background suppression.

Beyond CNN- and Transformer-based baselines, it is also useful to place our study in the context of earlier adversarial segmentation research. Authors [44] were to apply adversarial learning to semantic segmentation, where the discriminator promoted label-level consistency but lacked clear structural guidance. SegAN [45] built on this idea by introducing a multi-scale loss to capture both global and local dependencies. However, boundary precision was still not directly addressed, since edge priors and refinement signals were missing.

Pix2pix-cGAN [55] later proposed a conditional adversarial approach for medical image segmentation and has been used as a baseline in polyp segmentation studies. Our experiments show that its performance remains limited under a consistent evaluation setting. Even so, when our edge block and LRA are added, the results improve steadily, which suggests that these modules are not only central to our framework but can also benefit older adversarial models.

Inference-time analysis indicates that the model maintains tightly bounded latency, exhibiting less than $\pm 8.5\%$ deviation from the mean execution time. Such bounded variation is important in real-time systems. Meeting timing constraints and keeping high predictability are needed to avoid deadline violations and to ensure consistent frame processing. This characteristic enables practical deployment into clinical pipelines with strict frame-rate and responsiveness demands.

Fig. 5 shows typical failure cases where our model still encounters difficulties. Very small polyps may be missed because of weak visual cues, and severe occlusion caused by tissue folds or digestive fluid can hide important structures. Blurry or unclear boundaries often lead to incomplete masks, reflecting the challenge of separating subtle tissue transitions. In some images, high noise further lowers contrast and suppresses weak polyp signals. These limitations indicate that although our adversarial refinement improves performance, further work is still needed.

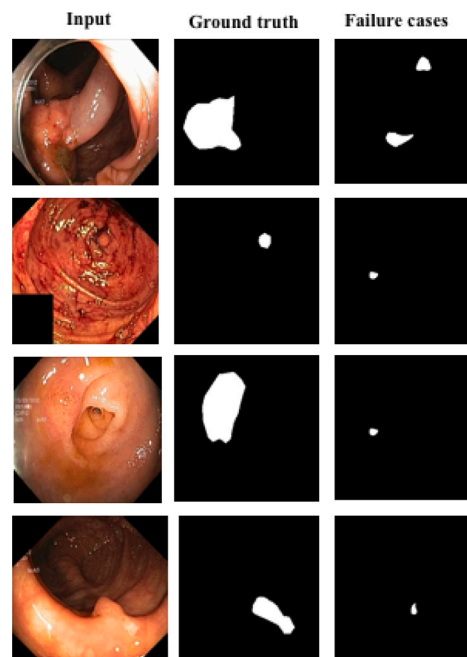


Fig. 5. Examples of failed cases illustrating segmentation limitations.

5. Conclusions and perspectives

This study presented an adversarial segmentation framework that integrates edge-guided priors and refinement-aware learning to enhance polyp boundary detection under constrained computational conditions. The method was evaluated on five benchmark datasets and several cross-domain settings. It achieved higher performance, stable inference speed, and strong generalization, which shows its suitability for clinical use. Ablation studies showed that the edge-guided module and the refinement attention module both contribute to better spatial accuracy and more stable temporal performance.

Future work may explore boundary-sensitive loss functions and adaptive resolution strategies for small polyps. Another direction is

lightweight domain adaptation to improve robustness in diverse and challenging clinical settings.

CRedit authorship contribution statement

Dinh Cong Nguyen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Duy Dat Tong:** Validation, Supervision, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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